

Charles Henville

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Education

University of Toronto

CGPA: 3.7/4.0 September 2022 - April 2027

Bachelor of Applied Science in Computer Engineering

Toronto, ON

Relevant Courses: Operating Systems, Algorithms & Data Structures, Digital Systems, Linear Algebra, Signals & Systems, Computer Organization, Software Communication & Design, Probability & Applications, Calculus I-III

Skills

Languages: Python, C++, C#, JavaScript, TypeScript, HTML/CSS, Java, SQL, Bash, Golang, Verilog, MATLAB

Technologies: Flask, React, Git, Linux, Docker, Kubernetes, AWS, Figma, PostgreSQL, Django, PyTorch, NumPy

Experience

Cerebras Systems

May 2025 - August 2026

Software Engineer Intern

Toronto, ON

- Designed and implemented **Python-based ETL** pipelines to consolidate cluster data (hardware specs, network configurations, etc.) from **10+** heterogeneous sources, reducing average manual data collection time by **95%**.
- Developed and introduced a new centralized data system, **leading** training sessions with **3+** teams to drive adoption and expand the scope of the data model, streamlining data accessibility for **25+** engineers.
- Designed and implemented telemetry dashboards with Figma, **React**, and **Django** alongside **Python** probing agents; enabling real-time observability for **8+** datacenters, generating alerts for **12+** engineers through automated pipelines.

Peripheral Labs, Inc.

December 2024 - May 2025

Founding Software Engineer

Toronto, ON

- Sole proprietor of a **React** and **WebGL/WebGPU**-based web application used to stream volumetric video for live events. Used as a technical PoC/MVP pitched to VCs and private investors, raising **\$2 million USD** in seed funding.
- Engineered a **Flask** backend responsible for encrypting, storing (**AWS s3**), and serving assets to construct the volumetric scene. Implemented client-side decryption post-auth with **Web Crypto API** preventing asset downloads.
- Developed a rendering pipeline for model-provided human pose estimations via algorithmic validation of the data and subsequent 3D object instantiation, positioning the company as a competitor to top organizations in sports analytics.

Toronto Police Service - Information Technology Services (ITS)

May 2024 - August 2024

Network and Data Engineer Intern

Toronto, ON

- Engineered a standalone Windows application with **Python/SQL** to facilitate inventory consolidation within ITS, resulting in data representing over **\$100 million** worth of inventory being consolidated into a single CMDB.
- Migrated a legacy **Drupal** Portal to a centralized CMS, improving workflow efficiency for **50+** employees.
- Configured or updated **10+** network devices (switches/routers) to be commissioned at **5+** police divisions.

Hubio Technology

June 2023 - September 2023

Data Engineer Intern

Toronto, ON

- Engineered efficient **Python** utility scripts for automatic and holistic comparison between remote **SQL** databases and **AWS s3** buckets, flagging and visualizing discrepancies in tables with upwards of **several million** records/entries.
- Designed and modified a custom data visualization/profiling application in **Python** with **Matplotlib**, **NumPy**, **Pandas**, and **SQL**, cumulatively highlighting **1000+** outlier and invalid records in client-provided **AWS s3** datasets.
- Implemented a predictive model using **PyTorch** and **Scikit-Learn** to profile and analyze legacy datasets for forecasting potential outlier frequency and future distributions; informing development decisions of **5+** analysts and engineers.

Leadership Experience

University of Toronto Creative Computing Association (Founder) 🔗

July 2024 - December 2025

- Founded a student association with **20+** members; developing student technical skills through interactive art displays.
- Led the development of a projects using **THREE.js**, **React**, **Flask** and **C++** firmware for microcontrollers.

Projects

Quantitative Equity Price Prediction through LSTMs, RNNs, and CNNs 🔗

April 2025

- Co-engineered neural networks & pipelines with **PyTorch** and **Pandas**; yielded a maximum **43.8%** annualized ROI.

Embedded Physics Engine and Peripheral Device Drivers 🔗

April 2024

- Designed a SPH-based fluid and rigid body physics engine in **embedded-C** for an ARM9 processor.